

31	27	26	25	24	20	19	15	14	12	11	7	6	0	
funct7				rs2		rs1		funct3		rd		opcode		R-type
rs3		funct2		rs2		rs1		funct3		rd		opcode		R4-type
imm[11:0]						rs1		funct3		rd		opcode		I-type
imm[11:5]				rs2		rs1		funct3		imm[4:0]		opcode		S-type

RV32D Standard Extension							
imm[11:0]			rs1	011	rd	0000111	FLD
imm[11:5]		rs2	rs1	011	imm[4:0]	0100111	FSD
rs3	01	rs2	rs1	rm	rd	1000011	FMADD.D
rs3	01	rs2	rs1	rm	rd	1000111	FMSUB.D
rs3	01	rs2	rs1	rm	rd	1001011	FNMSUB.D
rs3	01	rs2	rs1	rm	rd	1001111	FNMADD.D
0000001		rs2	rs1	rm	rd	1010011	FADD.D
0000101		rs2	rs1	rm	rd	1010011	FSUB.D
0001001		rs2	rs1	rm	rd	1010011	FMUL.D
0001101		rs2	rs1	rm	rd	1010011	FDIV.D
0101101		00000	rs1	rm	rd	1010011	FSQRT.D
0010001		rs2	rs1	000	rd	1010011	FSGNJ.D
0010001		rs2	rs1	001	rd	1010011	FSGNJN.D
0010001		rs2	rs1	010	rd	1010011	FSGNJX.D
0010101		rs2	rs1	000	rd	1010011	FMIN.D
0010101		rs2	rs1	001	rd	1010011	FMAX.D
0100000		00001	rs1	rm	rd	1010011	FCVT.S.D
0100001		00000	rs1	rm	rd	1010011	FCVT.D.S
1010001		rs2	rs1	010	rd	1010011	FEQ.D
1010001		rs2	rs1	001	rd	1010011	FLT.D
1010001		rs2	rs1	000	rd	1010011	FLE.D
1110001		00000	rs1	001	rd	1010011	FCLASS.D
1100001		00000	rs1	rm	rd	1010011	FCVT.W.D
1100001		00001	rs1	rm	rd	1010011	FCVT.WU.D
1101001		00000	rs1	rm	rd	1010011	FCVT.D.W
1101001		00001	rs1	rm	rd	1010011	FCVT.D.WU

RV64D Standard Extension (in addition to RV32D)													
1100001				00010	rs1	rm	rd	1010011		FCVT.L.D			
1100001				00011	rs1	rm	rd	1010011		FCVT.LU.D			
1110001				00000	rs1	000	rd	1010011		FMV.X.D			
1101001				00010	rs1	rm	rd	1010011		FCVT.D.L			
1101001				00011	rs1	rm	rd	1010011		FCVT.D.LU			

1111001	00000	rs1	000	rd	1010011	FMV.D.X
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